

Generated mesh
(GLB / OBJ / PLY)

Normalise to unit
bbox diagonal; weld
vertices; optional
QEM decimation to
10k faces (topology-
preserving)

Deterministic CPU probes
8 families, 30 features:
topology, boundary/manifold,
self-intersection, hidden
surface, thin wall, roughness,
triangle quality, size

Per-asset features
median 0.7-1.6 s
per mesh; peak RSS
 ≤ 345 MB; no GPU

Evaluated against:
expert defect labels
human geometry MOS
injected defects
MLLM pseudo-labels